

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B. Tech. Branch : Electronics and Computer Engineering/ Electronics and Computer Science

Semester : III Subject Code & Name: Electronics Devices and circuits(BTEPC302)

Max Marks: 60

Date: 07/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	A Field Effect transistor (FET) operates on a. Minority carriers only b. Majority carriers only c. Both majority & minority carriers d. None of these	CO1	1
2	At Pinch off voltage drain current for FET is a. Infinite b. Zero c. Both a & b d. None of these	CO1	1
3	The percentage voltage regulation for voltage supply providing 100v with no load and 95v at full load is a. 0.53% b. 5.26% c. 0.05% d. None of these	CO5	1
4	Which of the following circuit is known as emitter follower a. Zener shunt regulator b. Transistor shunt regulator c. Transistor series regulator d. Both a & b	CO5	1
5	Two stages of cascaded amplifier have gains of 50 & 20. Then overall voltage gain is a. 1000 b. 30 c. 70 d. 300	CO3	1
6	For impedance matching we can prefer	CO3	1

	a. RC coupled amplifier	b. Direct coupled amplifier	c. Transformer coupled amplifier	d. None of these		
7	An ideal opamp has				CO2	1
	a. Infinite A_v	b. Zero R_o	c. Infinite R_i	d. All the above		
8	Crystal oscillators are preferred than LC oscillators because of				CO3	1
	a. High Q-factor	b. High frequency stability	c. Small size of crystal	d. Both a & b		
9	The voltage gain of voltage follower is				CO2	1
	a. Less than unity	b. Equal to unity	c. Greater than unity	d. other than unity		
10	The drawback of negative feedback is				CO3	1
	a. Reduced gain	b. Frequency dependence	c. Temperature instability	d. None of these		
11	E-MOSFET is an				CO1	1
	a. Normally ON MOSFET	b. Normally OFF MOSFET	c. Normally ON/OFF MOSFET	d. None of these		
12	Which of the following is passive transducer				CO4	1
	a. Thermocouple	b. Piezoelectric transducer	c. Thermistor	d. photovoltaic cell		
Q. 2	Solve the following.					12
A)	Explain working of N-channel JFET with neat diagram.				CO1	6
B)	Draw and explain drain characteristics of N-MOSFET.				CO1	6
Q. 3	Solve the following.					12
A)	Draw and explain in detail the block diagram of Opamp.				CO2	6
B)	Explain Opamp as Non-inverting amplifier with suitable circuit diagram & waveforms.				CO2	6
Q. 4	Solve Any Two of the following.					12

A)	Explain construction and working of Hartley oscillator.	CO3	6
B)	Draw and explain RC-coupled amplifier.	CO3	6
C)	List different feedback topologies. Explain voltage series feedback connection with the help of block diagram.	CO3	6
Q. 5	Solve Any Two of the following.		12
A)	Explain three terminal positive voltage regulators.	CO5	6
B)	Explain the operation of transistor shunt regulator.	CO5	6
C)	Draw pin diagram of IC 723 and list its various features.	CO5	6
Q. 6	Solve Any Two of the following.		12
A)	Explain construction and working of LVDT.	CO4	6
B)	Draw construction and explain working of LDR.	CO4	6
C)	Explain construction and working of Thermocouple.	CO4	6
*** End ***			